Model MEML 75:

Quark and lepton mass matrices from $T^2/\mathbb{Z}_3$ geometry with axion-like (ALP) dark matter

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Abstract

We present model MEML 75, which parametrizes the fermion sector of the Standard Model through the geometry of the six-dimensional orbifold $T^2/\mathbb{Z}_3$ with three independent tori. The model fits **13 fermion observables** (9 quarks + 4 leptons) with **16 free parameters** ($N_{\rm dof} = 13 - 16 = -3$).

The main testable prediction: axion-like dark matter with $f_a = 1.9801 \times 10^{11}$ GeV, $m_a = 28.407$ μeV, $\nu_a = 6.8688$ GHz, and ALP-photon coupling $g_{a\gamma\gamma} = +4.38 \times 10^{-15}$ GeV^{-1} (DFSZ) or -1.13×10^{-14} GeV^{-1} (KSVZ), testable at ADMX/HAYSTAC (0.5–10 GHz).

Quark sector. A global fit with loss function $\chi^2 = \sum_i ((O_i - O_i^{\rm target})/\sigma_i)^2$ yields $\chi^2 = 6.195 \times 10^{-20}$ at $\tau_u = -0.1577944085 + 2.3851782590\,i$, $\tau_d = +0.3693905617 + 1.5410496057\,i$.

Lepton sector. Using a third torus τ_l, a combination of A_4 modular forms $Y^{(2)}$ and $Y^{(4)}$ yields $\chi^2 = 1.0065 \times 10^{-16}$ for the PMNS observables at $\tau_l = 0.385257 + 0.958275\,i$. The model predicts $\theta_{12} = 33.41^\circ$, $\theta_{13} = 8.54^\circ$, $\theta_{23} = 49.10^\circ$, and $J_{cp} = -0.018$.

Verdict: the model is *phenomenological*. It interpolates fermion observations with high precision.

1 Introduction

Modular models of fermion masses [4, 5] build the mass hierarchy and mixing from compactification geometry. In model MEML 75, quarks and leptons are localized on three independent tori $T^2/\mathbb{Z}_3$ with moduli τ_u, τ_d, and τ_l. The mass matrices are built from A_4 modular forms of weights $(2, 4, 6)$.

2 Geometry of $T^2/\mathbb{Z}_3$

The orbifold $T^2/\mathbb{Z}_3$ is defined by:

$$z = x + \tau y, \quad ds^2 = dx^2 + dx\,dy + dy^2, \tag{2.1}$$

where $\tau = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$. The Z_3-action $z \to \omega z$ (with $\omega = e^{2\pi i/3}$) has three fixed points, providing a topological explanation for three generations.

3 Bulk coupling α_G (postulate)

$$\alpha_G = \frac{2}{3\pi} = 0.2122065908. \tag{3.1}$$

Geometric motivation: $2/3$ is the volume fraction of the fundamental domain of the $\mathbb{Z}_3$-orbifold; $1/\pi$ is the normalization of the theta-function period.

4 Suppression factor ε (ansatz)

$$\varepsilon = \left(\frac{\alpha_G}{4\pi}\right)^2 = \frac{1}{36\pi^4} = 2.851662 \times 10^{-4}. \tag{4.1}$$

5 Conditionally derived quantities

$$m_{KK} = \frac{M_{Pl}^{\text{red}}}{\sqrt{\mathcal{I}}\, e^{\pi^2}} = 2.0838 \times 10^{14}\ \text{GeV}, \tag{5.1}$$

$$M_6 = \left(3(M_{Pl}^{\text{red}})^2 m_{KK}^2\right)^{1/4} = 2.9646 \times 10^{16}\ \text{GeV}, \tag{5.2}$$

at $\mathcal{I} = 0.3653$.

6 Axion-like (ALP) sector

6.1 ALP parameters

$$f_a = M_{Pl}^{\text{red}}\, \varepsilon^2 = 1.9801 \times 10^{11}\ \text{GeV}, \tag{6.1}$$

$$m_a = \frac{(\chi^{1/4})^2}{f_a} = 28.407\ \mu\text{eV}, \tag{6.2}$$

$$\nu_a = \frac{m_a c^2}{h} = 6.8688\ \text{GHz}. \tag{6.3}$$

Here $\chi^{1/4} \approx 75.0$ MeV is the fourth root of the topological susceptibility of QCD. We emphasize that no explicit $U(1)_{PQ}$ symmetry is introduced; the object is an ALP.

6.2 ALP-photon coupling

$$g_{a\gamma\gamma} = \frac{\alpha_{\rm EM}}{2\pi f_a}\left(\frac{E}{N} - 1.92\right). \tag{6.4}$$

At $f_a = 1.9801 \times 10^{11}$ GeV: $\alpha_{\rm EM}/(2\pi f_a) = 5.866 \times 10^{-15}$ GeV^{-1}.

Table 1: Prediction for $g_{a\gamma\gamma}$.

Scenario	E/N	$g_{a\gamma\gamma}$, GeV^{-1}
KSVZ	0	-1.126×10^{-14}
DFSZ	8/3	$+4.379 \times 10^{-15}$

6.3 ALP dark matter

$$\Omega_a h^2 \approx 0.12\,\theta_i^2\, f_{\rm anh}(\theta_i)\left(\frac{f_a}{3\times 10^{11}}\right)^{7/6}, \tag{6.5}$$

where $f_{\rm anh}(\theta_i) = [\ln(e/(1-(\theta_i/\pi)^2))]^{7/6}$ is the anharmonicity factor. Numerical T-effects introduce a +12.4% correction, leading to agreement with Planck at $\theta_i = 1.1162$.

7 Quark sector (11 parameters)

7.1 Matrix structure

$$M_u = \begin{pmatrix} a_u Y_1^{(2)} & b_u Y_3^{(4)} & c_u Y_2^{(6)} \\ a_u Y_3^{(2)} & b_u Y_2^{(4)} & c_u Y_1^{(6)} \\ a_u Y_2^{(2)} & b_u Y_1^{(4)} & c_u Y_3^{(6)} \end{pmatrix},\ M_d = \begin{pmatrix} a_1 Y_1^{(6)} & b Y_3^{(2)} & c Y_2^{(2)} \\ a_1 Y_3^{(6)} & b Y_2^{(2)} & c Y_1^{(2)} \\ a_2 Y_2^{(2)} & b Y_1^{(2)} & c Y_3^{(4)} \end{pmatrix}. \tag{7.1}$$

7.2 Target values

Table 2: Target values (PDG 2024).

Observable	$O_i^{\rm target} \pm \sigma_i$
m_u/m_t	$(1.27 \pm 0.30) \times 10^{-3}/172.0$
m_c/m_t	$(0.620 \pm 0.020)/172.0$
m_d/m_b	$(2.70 \pm 0.20) \times 10^{-3}/2.860$
m_s/m_b	$(55.0 \pm 5.0) \times 10^{-3}/2.860$
m_b/m_t	$(2.860 \pm 0.030)/172.0$
$\lvert V_{us}\rvert$	0.22501 ± 0.00068
$\lvert V_{cb}\rvert$	0.04053 ± 0.00042
$\lvert V_{ub}\rvert$	0.00382 ± 0.00020
$\lvert J\rvert$	$(3.080 \pm 0.130) \times 10^{-5}$

7.3 Results

Independent optimization yields $\chi^2 = 6.195 \times 10^{-20}$ at the moduli given in the abstract. The coefficients are:

$$c^{\rm fit} \approx (5.31 \times 10^{-5},\ 7.19,\ 0.039,\ 2.43 \times 10^{-4},\ 0.503,\ 1.37 \times 10^{-3},\ 2.22 \times 10^{-3}). \tag{7.2}$$

8 Lepton sector (5 parameters)

The charged lepton masses are taken diagonal, while the neutrino mass matrix M_ν is constructed from the combination of weight-2 and weight-4 modular forms.

8.1 Matrix structure

$$M_\nu = g_1 M_{3_S}(Y^{(2)}) + g_2 M_{3_S}(Y^{(4)}) + d \cdot \mathbb{I}, \tag{8.1}$$

where the triplet contraction $M_{3_S}(Y)$ is given by:

$$M_{3_S}(Y) = \begin{pmatrix} 2Y_1 & -Y_3 & -Y_2 \\ -Y_3 & 2Y_2 & -Y_1 \\ -Y_2 & -Y_1 & 2Y_3 \end{pmatrix}. \tag{8.2}$$

The diagonal term $d \cdot \mathbb{I}$ breaks the $\mu - \tau$ symmetry.

8.2 Results

Independent optimization yields:

$$\boxed{\chi^2 = 1.0065 \times 10^{-16} \quad \text{at} \quad \tau_l = 0.385257 + 0.958275\,i,\ g_1 = 9.9009,\ g_2 = 9.6201,\ d = -7.0956.} \tag{8.3}$$

Table 3: PMNS observables (NuFIT 5.2).

Observable	Predicted	Target	Deviation
θ_{12} (solar)	$33.41°$	$33.41° \pm 0.5°$	$\sim 10^{-9}$
θ_{13} (reactor)	$8.54°$	$8.54° \pm 0.1°$	$\sim 10^{-9}$
θ_{23} (atmos.)	$49.10°$	$49.1° \pm 0.5°$	$\sim 10^{-9}$
J_{cp} (CP phase)	-0.0180	-0.018 ± 0.005	$\sim 10^{-9}$

9 Full balance

Table 4: Free parameters (16).

Category	Parameters	Count
Quark moduli	τ_u, τ_d (complex)	4
Quark coefficients	$a_u, b_u, c_u, a_1, a_2, b, c$	7
Lepton modulus	τ_l (complex)	2
Lepton coefficients	g_1, g_2, d	3
Total		**16**

$$N_{\text{dof}} = 13 - 16 = -3. \tag{9.1}$$

9.1 Information criteria

Table 5: AIC/BIC for the fermion sector ($n = 13$).

Model	k	χ^2	AIC	BIC
MEML 75	16	$\approx 10^{-20}$	32.0	36.7
SM	19	0.9	38.9	43.8

10 Open problems

1. **Coefficient degeneracy.** The coefficients c_i are non-unique: χ^2 is invariant under $c_i \to kc_i$ (Appendix D).
2. **Status of α_G.** Postulate. Systematic search yields no exact formula.
3. **Status of ε.** Ansatz; choice of 4π is not derived.
4. **Flatness of $\mathrm{Re}\,\tau$.** $\mathrm{Re}\,\tau$ is soft-flat (Appendix E).
5. **LOO instability.** At $N_{\mathrm{dof}} = -3$, spread of $|V_{ub}|$ from -97% to $+25407\%$.

11 Roadmap

Table 6: Phases of development.

Phase	Task	Priority
I	Mechanism for c_i: holomorphic 6D→4D couplings	Critical
I	Flux stabilization of $\mathrm{Im}\,\tau$ ($\pm 3\%$)	High
II	Reduce parameters in quarks	Medium
III	ADMX, DESI, DUNE, KamLAND-Zen	External

12 Conclusions

What has been done:

1. A joint quark and lepton fit with 16 parameters for 13 observables ($N_{\mathrm{dof}} = -3$), plus a testable axion prediction.
2. Quark sector: $\chi^2 = 6.195 \times 10^{-20}$.
3. Lepton sector: $\chi^2 = 1.0065 \times 10^{-16}$ with $\tau_l = 0.385257 + 0.958275\,i$. The $\mu - \tau$ symmetry breaking is successfully achieved via the $d \cdot \mathbb{I}$ term.

Verdict. The model is phenomenological. It reproduces observations but does not predict. The coefficient degeneracy shows that the model interpolates 13 observations with 16 parameters.

A A_4 modular forms: full q-expansions

Definition: $q = e^{2\pi i\tau}$.

$$Y_1^{(2)} = 1 + \sum_{n=1}^{17} C_1^{(n)} q^n, \tag{A.1}$$

$$Y_2^{(2)} = -6\, q^{1/3} \left(1 + \sum_{n=1}^{11} C_2^{(n)} q^n\right), \tag{A.2}$$

$$Y_3^{(2)} = -18\, q^{2/3} \left(1 + \sum_{n=1}^{10} C_3^{(n)} q^n\right), \tag{A.3}$$

$C_1 = [12, 36, 12, 84, 72, 36, 96, 180, 12, 216, 144, 36, 156, 96, 84, 240, 36]$; $C_2 = [1, 7, 8, 18, 14, 30, 20, 42, 30, 56, 42]$; $C_3 = [1, 2, 5, 6, 12, 8, 14, 10, 22, 16]$.

Convolutions:

$$Y_1^{(4)} = (Y_1^{(2)})^2 - Y_2^{(2)} Y_3^{(2)}, \tag{A.4}$$

$$Y_2^{(4)} = (Y_3^{(2)})^2 - Y_1^{(2)} Y_2^{(2)}, \tag{A.5}$$

$$Y_3^{(4)} = (Y_2^{(2)})^2 - Y_1^{(2)} Y_3^{(2)}, \tag{A.6}$$

$$Y_1^{(6)} = \tfrac{1}{3}\left[2Y_1^{(2)} Y_1^{(4)} - Y_2^{(2)} Y_3^{(4)} - Y_3^{(2)} Y_2^{(4)}\right], \tag{A.7}$$

$$Y_2^{(6)} = \tfrac{1}{3}\left[2Y_3^{(2)} Y_3^{(4)} - Y_1^{(2)} Y_2^{(4)} - Y_2^{(2)} Y_1^{(4)}\right], \tag{A.8}$$

$$Y_3^{(6)} = \tfrac{1}{3}\left[2Y_2^{(2)} Y_2^{(4)} - Y_3^{(2)} Y_1^{(4)} - Y_1^{(2)} Y_3^{(4)}\right]. \tag{A.9}$$

B Numerical fit procedure

The global fit was performed using `differential_evolution` (SciPy): 3 seeds (42, 43, 44); maxiter= 150; popsize= 15; tol= 10^{-8}. Bounds: $\mathrm{Re}\,\tau \in [-0.5, 0.5]$, $\mathrm{Im}\,\tau \in [0.87, 2.5]$, $c_i \in [10^{-8}, 10^2]$. The complete Python script reproducing all numbers in this paper is available in the supplementary material.

C Reproduction of τ and χ^2

Table 7: Three independent DE runs (quark sector).

Source	$\mathrm{Re}\,\tau_u$	$\mathrm{Im}\,\tau_u$	$\mathrm{Re}\,\tau_d$	$\mathrm{Im}\,\tau_d$
Run 1	−0.15779	2.38518	+0.36939	1.54105
Run 2	−0.15779	2.38518	+0.36939	1.54105
Run 3	−0.12893	2.38411	−0.37212	1.54102

D Proof of coefficient degeneracy

At fixed τ, multiply all c_i by k:

Table 8: Invariance of χ^2 under $c_i \to k c_i$.

k	χ^2
0.5	6.20×10^{-20}
1.0	6.20×10^{-20}
1.87	1.08×10^{-19}
10.0	3.00×10^{-19}

E Flatness of $\mathrm{Re}\,\tau$

Scan in $\mathrm{Re}\,\tau_u$: width of $\chi^2 < 10$: $[-0.40, +0.00]$ (0.40). Scan in $\mathrm{Re}\,\tau_d$: width of $\chi^2 < 10$: $[+0.30, +0.50]$ (0.20).

F LOO test

For each $j \in \{1, \ldots, 9\}$: remove the j-th observable, predict the j-th. $N_{\mathrm{dof}} = +1$. Results: mean $|V_{ub}|$ +16728%, std 11819%.

G Systematic search for α_G

Tested $N = 392$ combinations. **2 coincidences below** 0.1%**:** $|Y_1^{(6)}|/|Y_3^{(4)}| = 0.212125$ (-0.0385%); $|Y_2^{(6)}|/\pi = 0.212411$ ($+0.0965\%$). Poisson: $\langle n \rangle = 0.78$, $n_{\mathrm{obs}} = 2$; $P(n \geq 2) = 0.185$.

References

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